

interfaces. It basically runs on commodity hardware. It was developed by Thomas d'Otrepe de Bouvette, the creator of Aircrack software. OpenWIPS uses many functions and services built into Aircrack-NG for scanning, detection and intrusion prevention.

The three main parts of OpenWIPS-NG are listed below.

Sensor: Acts as a device for capturing wireless traffic and sending the data back to the server for further analysis. The sensor also plays an important role in responding to all sorts of network attacks.

Server: Performs the role of aggregation of data from all sensors, analyses the data and responds to attacks. Additionally, it logs any type of attack and alerts the administrator.

Interface: The GUI manages the server and displays the information regarding all sorts of threats against the network.

Pros

- Modular and plugin based.
- Software and hardware required can be built by DIYers.
- Additional features are supported via use of plugins.

Cons

- Only works for wireless networks.
- Only suitable for low and medium level administration, and not fully compliant for detecting all sorts of wireless attacks.
- No detailed documentation and community support compared to other systems.

Latest version: OpenWIPS-NG 0.1 beta 1

Official website: <http://www.openwips-ng.org/>

Suricata

Suricata is an open source, fast and highly robust network intrusion detection system developed by the Open Information Security Foundation. The Suricata engine is capable of real-time intrusion detection, inline intrusion prevention and network security monitoring. Suricata consists of a few modules like Capturing, Collection, Decoding, Detection and Output. It captures traffic passing in one flow before decoding, which is highly optimal. But unlike Snort, it configures separate flows after capturing and specifying how the flow will separate between processors.

Pros:

- Does the network traffic processing on the seventh layer of the OSI model which, in turn, enhances its capability to detect malware activities.
- Automatically detects and parses protocols like IP, TCP, UDP, ICMP, HTTP, TLS, FTP, SMB and FTP so that rules apply on all protocols.
- Advanced features consist of multi-threading and GPU acceleration.

Cons:

- Less support as compared to other IDSs like Snort.
- Complicated in operation and requires more system resources for full-fledged functioning.

Latest version: 3.2

Official website: <https://suricata-ids.org>

BroIDS

BroIDS is a passive, open source network traffic analyser developed by Vern Paxson, and is used for collecting network measurements, conducting forensic investigations, traffic base lining and much more. BroIDS comprises a set of log files to record network activities like HTTP sessions with URIs, key headers, MIME types, server responses, DNS requests, SSL certificates, SMTP sessions, etc. In addition, it provides sophisticated functionality for the analysis and detection of threats, extracting files from HTTP sessions, sophisticated malware detection, software vulnerabilities, SSH brute force attacks and validating SSL certificate chains.

BroIDS is divided into the following two layers.

Bro Event Engine: This does the task of analysing live or recorded network traffic packs using C++ to generate events when something unusual happens on the network.

Bro Policy Scripts: These analyse events to create policies for action, and events are handled using policy scripts such as sending emails, raising alerts, executing system commands and even calling emergency numbers.

Latest version: Bro 2.5

Official website: www.bro.org

Pros:

- Highly flexible as BroIDS uses a scripting language to allow users to set monitoring rules for each protected object.
- Works efficiently in networks with large volumes of traffic and handles big network projects.
- Capable of in-depth analysis of traffic and supports analysers for multiple protocols. Highly stateful and does forensic level comprehensive log maintenance.

Cons:

- Not easy to handle as it has a complex architecture.
- Programming experience is required for competent handling of the BroIDS system.

OSSEC

OSSEC is a free and open source host based IDS that performs varied tasks like log analysis, integrity checking, Windows registry monitoring, rootkit detection, time-based alerting and active response. The OSSEC system is equipped with a centralised and cross-platform architecture allowing multiple systems to be accurately monitored by administrators.